COM-480 Data visualization

Milestone 1

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1 Dataset

Our project mainly uses the *World Development Indicators* (WDI) dataset[1] published by the World Bank. WDI is a widely used public dataset that contains more than 1,000 economic and social indicators for over 200 countries from 1960 to the present. It is commonly used in economics and development research and therefore provides reliable and well-documented data.

In addition to the WDI dataset, we use several external information sources listing countries participating in the Belt and Road Initiative (BRI)[2] and regional information[3]. This allows us to classify countries into BRI and non-BRI groups and compare their development trajectories, as well as the regions they belong to. We initially considered using official data from the BRI website[4]; however, due to restrictions on the number of downloadable entries and other access limitations, we used it only as a supplementary source to fill in missing values.

The WDI dataset is clean and well-structured for most of the features and is available for download in CSV format. Data are provided in tabular format with official country names and yearly observations, which facilitates merging and comparison across indicators. However, some preprocessing is required. Several indicators contain missing values, particularly for earlier and epidemic years or for smaller economies. To address this issue, we will restrict the analysis to the period after 2000 and focus on specific countries that represent the region and categories.

Finally, the dataset will be reshaped from wide format (with years as columns) into a long time-series format in order to facilitate visualization and analysis. Overall, the dataset is of high quality and requires only moderate preprocessing, making it well-suited for a data visualization project.

2 Problematic

Large international cooperation initiatives often claim to promote economic development and regional connectivity. One of the most prominent examples is the Belt and Road Initiative (BRI), launched in 2013, which aims to strengthen infrastructure networks, trade links, and economic collaboration across Asia, Europe, and Africa. However, it remains unclear whether countries participating in such initiatives actually experience different development trajectories compared to similar countries that are not part of these frameworks.

The goal of this project is to explore whether countries involved in the BRI show observable differences in development patterns compared to non-BRI countries. Rather than focusing on a single economic indicator, we aim to investigate multiple dimensions of development, including the aspects of economic growth, demographic changes, technological adoption, and government priorities. By combining indicators such as GDP per capita, technology access, trasportation development, trade, and many more, we seek to visualize how countries evolve over time and whether these trajectories differ between BRI and non-BRI groups.

Our motivation is to provide an exploratory and data-driven perspective on global development patterns. Public discussions about international cooperation initiatives often emphasize their economic impact, yet it is difficult to assess these claims without examining long-term data across multiple indicators, and there is a much broader scope for the measurement of the development of the countries beyond the economy.

The target audience includes students, researchers, and readers interested in global development, international cooperation, and data-driven storytelling. Through interactive visualizations, the project aims to help users better understand similarities and differences in development trajectories across countries and highlight broader patterns in how societies transform over time.

3 Exploratory Data Analysis

Our EDA assesses data quality and uncovers baseline trends regarding the Belt and Road Initiative (BRI). We reshaped the raw World Bank wide-format data into two complementary structures: a **long-format panel dataset** (*Country-Year-Indicator*) and a **time-series dataset**. Maintaining both versions optimizes our analytical workflow: the panel data facilitates cross-sectional baseline comparisons, while the time-series format streamlines the rendering of longitudinal interactive visualizations.

[1] https://databank.worldbank.org/source/world-development-indicators
[2] https://en.wikipedia.org/wiki/Belt_and_Road_Initiative
[3] https://greenfdc.org/countries-of-the-belt-and-road-initiative-bri/
[4] https://www.yidaiyilu.gov.cn/dataChart

3.1 Data Quality and Imputation

Macroeconomic data inherently contains missing values (Figure 1). Variables with > 80% missingness were excluded to maintain analytical integrity. For intermittent gaps in core indicators, we applied within-country linear interpolation (max 2 consecutive years) to preserve visual continuity without synthesizing artificial trends.

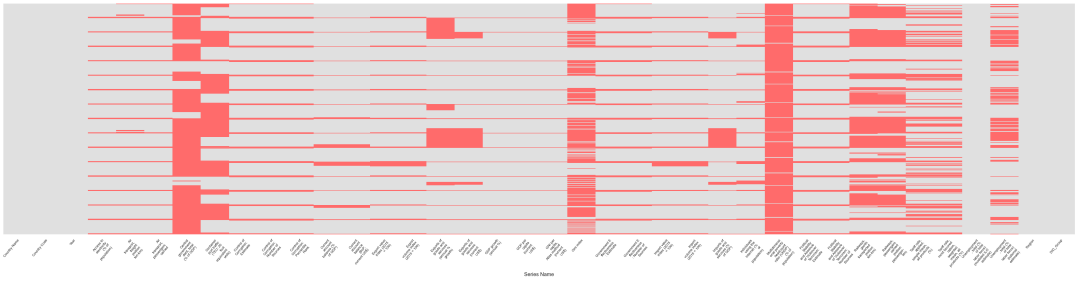

Figure 1: Missing value heatmap (grey: present, red: missing). Variables with > 80% missingness were excluded.

3.2 Visualization Strategy & Temporal Trends

Although we curated dozens of metrics from the WDI, our EDA visualizations highlight specific proxies to anchor the narrative. We explicitly plotted *GDP per capita* (Figure 2) as the ultimate macroscopic measure of economic health, aiming to observe if top-level growth divergence emerged between BRI and non-BRI cohorts post-2013. To frame this, we categorized countries into five BRI regional subsets versus a matched non-BRI control group (India, Viet Nam, Tanzania), with China as the source. Post-2013 trajectories reveal heterogeneous macroeconomic patterns requiring interactive exploration.

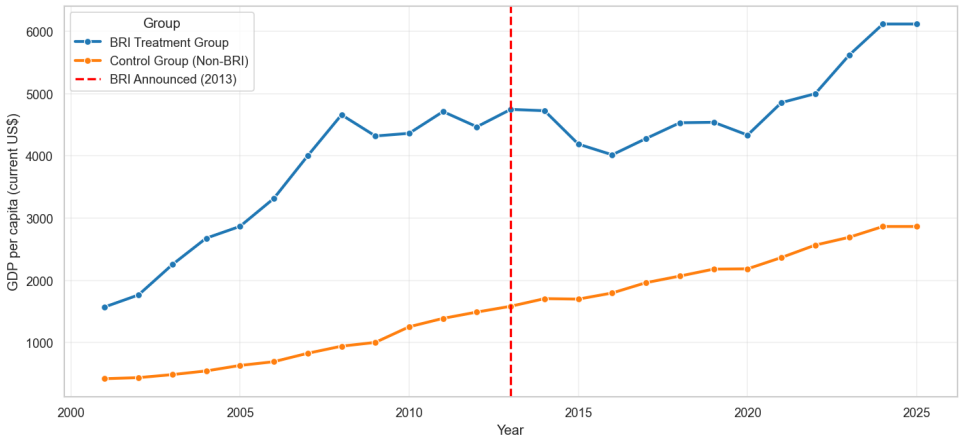

Figure 2: GDP per capita trends. The red dashed line denotes the 2013 BRI announcement.

3.3 Baseline Disparities & Synergy

To isolate structural differences from COVID-19 anomalies, we set 2019 as a cross-sectional baseline. Rather than economic metrics, we plotted institutional (*Control of Corruption*) and infrastructural indices via grouped boxplots (Figure 3) to expose significant non-financial baseline disparities, particularly between African/South Asian subsets and controls.

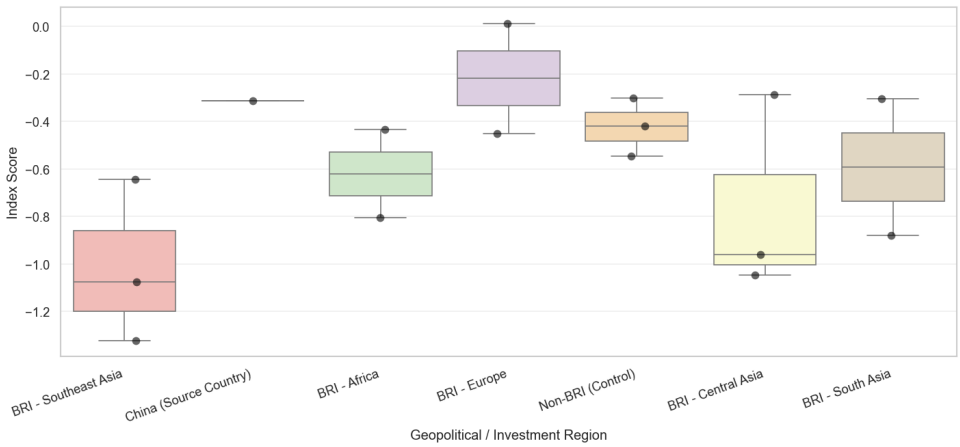

Figure 3: Regional distribution of institutional and infrastructural indices (2019 baseline).

Finally, to justify our multi-dimensional approach, a correlation matrix (Figure 4) was plotted to validate theoretical synergies. It confirms strong positive links between transportation infrastructure and trade connectivity, while revealing complex governance relationships that will drive our final dashboard's narrative.

Figure 4: Correlation matrix of selected key indicators validating multidimensional synergies.

4 Related Work

The World Development Indicators (WDI) dataset has been extensively used for visualizing global development patterns, most notably through tools such as Gapminder's Trendalyzer[5] , which enables dynamic exploration of relationships between variables like GDP per capita and life expectancy over time. Similar approaches have combined WDI with governance or development indices to produce comparative scatterplots and dashboards that highlight broad cross-country differences[6]. These works primarily focus on general development trajectories and correlations across countries, often without emphasizing specific policy frameworks or interventions.

Our approach diverges in two key ways. First, we adopt a quasi-experimental perspective by explicitly structuring the dataset into a treatment group (BRI countries) and a matched control group (non-BRI countries). Rather than presenting development as a continuous global spectrum,

[5]https://www.gapminder.org/tools-offline/
[6]https://www.dimiter.eu/Visualizations_files/WDVP.html

we aim to isolate and visually compare trajectories between these groups, particularly before and after the 2013 BRI announcement. This framing introduces a causal interpretation layer that is largely absent from existing visualization-focused work using WDI data.

Second, our focus on the Belt and Road Initiative represents a relatively underexplored topic in the context of interactive data visualization. While prior studies have examined BRI-related indicators such as trade, FDI, or emissions[7] , these analyses are typically static and aggregate BRI countries into a single unit. In contrast, our visualization would likely emphasize individual country trajectories within and across groups, allowing for direct comparison and the identification of heterogeneous patterns even between fellow BRI countries.

Inspired by dynamic visualization tools like Gapminder, our project extends this paradigm by embedding a policy-centered, comparative narrative. The goal is not only to observe how countries evolve, but to explore whether participation in a major international initiative corresponds to distinguishable developmental pathways.

[7] ResearchGate figure